

Non-Data-Aided SDM MIMO Equalization for Randomly-Coupled Multi-Core Fibers with a Tap-Correlation Avoidance CMA

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Abstract We propose the tap-correlation avoidance (TCA-)CMA, a low-complexity, non-data-aided MIMO equalizer for RC-MCFs, preventing singularities by penalizing tap-weight correlations. Simulations using 32 GBd 16-QAM signals show singularity-free convergence up to 600 ps total delay spread for 3-, 4-core fibers and 300 ps for 7-core fiber. ©2026 The Author(s)

Introduction

Randomly-coupled multi-core fibers (RC-MCFs) enable Pb/s long-haul transmission [1] through high spatial density and low total delay spread [2], but require complex high-order multiple-input multiple-output (MIMO) equalization [3]. A well-known non-data-aided equalizer, the constant modulus algorithm (CMA) [4], offers low complexity and robustness to phase and frequency noise but may converge to identical outputs or delayed replicas (*singularities*), especially with many channels [5] or large spatial mode dispersion (SMD) [6].

Previously, we experimentally demonstrated CMA extensions for singularity-free, non-data-aided MIMO equalization in SDM transmissions: first, over 3-mode few-mode fiber (FMF) by incorporating a cost function that penalizes correlations between the equalizer's outputs [7]; and second, over 3-core RC-MCF by orthonormalizing the equalizer's tap weights across the total delay spread between successive CMA multiple-input, single-output (MISO) convergences [8]. In this work, we propose the tap-correlation avoidance (TCA-)CMA, which utilizes a cost function that penalizes correlations among the equalizer's tap weights instead of the outputs. Output-based singularity detection may be more sensitive to noise, necessitating averaging. In contrast, tap weights are less noisy, removing the need for averaging, and because tap weights do not change significantly from symbol to symbol, correlations can be computed less frequently. These factors make

the TCA-CMA more computationally efficient and particularly attractive for real-time implementation.

Here, we compare a 39-tap TCA-CMA for dual polarization (DP)-16-QAM 32 GBd transmission over 3-, 4-, and 7-core RC-MCFs at various distances using a multi-section model [9] (1800 realizations in total) with the conventional CMA, both used as pre-convergence for a multi-modulus algorithm (MMA). We observed singularity-free convergence for 3- and 4-core RC-MCFs up to 600 ps total delay spread and for 7-core RC-MCFs up to 300 ps total delay spread, while the conventional CMA failed in 98% of cases. Additionally, we assessed the performance of the TCA-CMA at various noise levels, demonstrating low penalty for 3-, 4-, and 7-core RC-MCFs.

Multi-Section Model for RC-MCF Transmission

To investigate the impact of SMD, core count, and transmission distance on the proposed non-data-aided MIMO equalizer, we employ the multi-section model from [9], as illustrated in Fig. 1(a). In this model, the fiber is divided into S sections, and the frequency response of section s of an N -core fiber is described by a $2N \times 2N$ matrix $\mathbf{M}_s(\omega)$. Cascading all S sections yields the overall channel matrix: $\mathbf{M}(\omega) = \mathbf{M}_S(\omega) \cdots \mathbf{M}_2(\omega) \mathbf{M}_1(\omega)$. Each section is modeled as the product of three $2N \times 2N$ matrices: $\mathbf{M}_s(\omega) = \mathbf{V}_s \mathbf{\Lambda}_s(\omega) \mathbf{U}_s^H$, where $\mathbf{U}_s$ and $\mathbf{V}_s$ are random unitary matrices representing the coupling at the section input and output, respectively, and H denotes the Her-

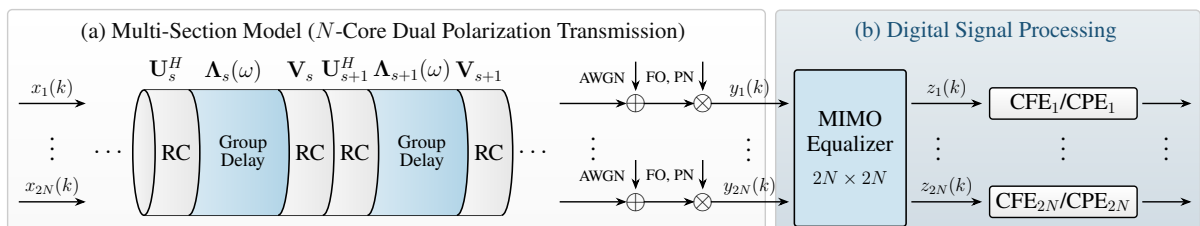


Fig. 1: (a) Multi-section model for N -core DP transmission: unitary matrices $\mathbf{U}$, $\mathbf{V}$ represent random coupling (RC), and diagonal matrix $\mathbf{\Lambda}$ represents group delay. AWGN: additive white Gaussian noise, FO: frequency offset, PN: phase noise. (b) Digital signal processing including the $2N \times 2N$ non-data-aided MIMO equalizer and carrier frequency/phase estimation (CFE/CPE).

mitian transpose. The diagonal matrix $\Lambda_s(\omega)$ describes the group delays (GDs): $\Lambda_s(\omega) = \text{diag}[e^{-j\omega\tau_{s,1}}, e^{-j\omega\tau_{s,2}}, \dots, e^{-j\omega\tau_{s,2N}}]$, where $\tau_{s,i}$ with $i \in \{1, \dots, 2N\}$ are the per-section zero-mean GDs with standard deviation σ_τ . For our study, we neglect chromatic dispersion and mode-dependent loss/gain. After transmission simulation, the signals are impaired by additive white Gaussian noise (AWGN) at various SNR levels, a frequency offset (FO) of 100 MHz, and phase noise (PN) modeled as a Wiener process with a 100 kHz linewidth. As displayed in Fig. 1(b), the received signals, at 2 samples per symbol (sps), are processed by the $2N \times 2N$ non-data-aided MIMO equalizer (detailed in the next section), followed by carrier frequency/phase estimation (CFE/CPE) using conventional methods for coherent digital signal processing (DSP) [10].

TCA-CMA for SDM Transmission

For MIMO equalization of a DP N -core SDM transmission, for $i, j \in \{1, \dots, 2N\}$, the j -th MIMO output symbol sequence is computed as $z_j(k) = \sum_{i=1}^{2N} \mathbf{h}_{ij}^T(k) \mathbf{y}_i(k)$, where T denotes vector transposition, $\mathbf{h}_{ij}(k)$ is a column vector containing N_t tap weights, and $\mathbf{y}_i(k)$ is a column vector containing N_t time-reversed samples of the i -th equalizer input. The tap weights are updated via gradient descent to minimize the following TCA-CMA cost function:

$$J_{\text{TCA-CMA}}(k) = J_{\text{CMA}}(k) + \alpha J_{\text{TCA}}(k) \quad (1)$$

where $J_{\text{CMA}}(k)$ corresponds to the conventional CMA cost function [4]:

$$J_{\text{CMA}}(k) = \sum_{j=1}^{2N} \left(|z_j(k)|^2 - R_2 \right)^2 \quad (2)$$

with the CMA constant $R_2 = \mathbb{E}[|a|^4] / \mathbb{E}[|a|^2]$ computed over the constellation symbols a . Here, $J_{\text{TCA}}(k)$ is the newly introduced tap-correlation avoidance cost function to prevent singularities by detecting correlations between the equalizer's tap weights:

$$J_{\text{TCA}}(k) = 2 \sum_{\substack{j,n=1 \\ n \neq j}}^{2N} \sum_{\substack{\delta=-N_t+1 \\ \delta=N_t-1}}^{N_t-1} |r_{jn,\delta}(k)|^2 \quad (3)$$

where $r_{jn,\delta}(k) = \sum_{i=1}^{2N} \mathbf{h}_{ij}^T(k) \mathbf{h}_{in,\delta}^*(k)$ is the tap-correlation between the tap-weight vectors of outputs j and n with the same input i , with a tap-shift δ applied to $\mathbf{h}_{in}$, and $*$ denotes complex conjugation. Since singularities may appear as delayed replicas, correlation evaluation at multiple tap delays is required, as previously shown in [8]. Therefore, the correlation between tap-weight vectors is computed for all shifts δ across the full tap range, summing only where $\mathbf{h}_{ij}$ and the shifted $\mathbf{h}_{in,\delta}$ overlap. The resulting tap-weight update equation is:

$$\begin{aligned} \mathbf{h}_{ij}(k+1) = \mathbf{h}_{ij}(k) - 2\mu & \underbrace{\left[(|z_j(k)|^2 - R_2) z_j(k) \mathbf{y}_i^*(k) \right]}_{\text{CMA term}} \\ & + \underbrace{\alpha \sum_{\substack{n=1 \\ n \neq j}}^{2N} \sum_{\delta=-N_t+1}^{N_t-1} r_{jn,\delta}(k) \mathbf{h}_{in,\delta}(k)}_{\text{tap-correlation penalty}} \end{aligned} \quad (4)$$

where $\mu > 0$ is the step size and $\alpha \geq 0$ weights the tap-correlation penalty.

Unlike our previous approach [7], which penalizes correlations among equalizer outputs and requires temporal averaging, the proposed method computes correlations directly from tap weights, which are less noisy and thus require no averaging. Furthermore, because tap weights do not change significantly from symbol to symbol, correlations can be computed every K symbols, reducing memory and complexity.

Results and Discussion

To evaluate the singularity-free TCA-CMA convergence for MIMO equalization in RC-MCF systems, we simulated 1800 cases across three core counts (3-, 4-, 7-cores), four transmission distances (100, 225, 400, 900 km), and three SMD values (3, 4, 5 ps/ $\sqrt{\text{km}}$), with 50 realizations per configuration. The SMD range aligns with reported RC-MCF measurements, including a record-low of 3.14 ps/ $\sqrt{\text{km}}$ for 4-core fiber [2, 11, 12]. Although current 7-core RC-MCFs have higher SMD than simulated here, manufacturing advances are expected to reduce these values, as demonstrated for fibers with even higher core counts [2, 12]. In the multi-section model, strong coupling can be assumed if the number of sections S is large. Under this assumption, we set each section length to 1 km, so the number of sections S equals the fiber length in km and $\sigma_\tau = \sigma_{\text{km}}$ corresponds to the SMD in ps/ $\sqrt{\text{km}}$ [9].

We employed a 39-tap TCA-CMA, i.e. a low tap count to ease non-data-aided convergence, as a pre-convergence stage for DP-16-QAM signals at 32 GBd, yielding an equalizer length of approximately 585 ps (at 2 sps). Distances were chosen with integer square roots to ease comparison of the total delay spread, which approaches $\pm 2\sigma_\tau \sqrt{S}$ for statistically independent and identical sections [13]. To limit the simulation length to 3.2×10^5 symbols, we ran the 39-tap TCA-CMA ($\mu = 10^{-4}$, $\alpha = 4$) for 5 pre-convergence iterations, chosen conservatively to enable comparison across all configurations. The tap-correlation penalty in (4) was only updated every $K = 1000$ symbols. Subsequently, the equalizer switched to a 101-tap MMA ($\mu = 2 \cdot 10^{-5}$) for improved performance on 16-QAM signals [14].

Fig. 2 compares the convergence outcomes of the $2N \times 2N$ MIMO equalizer using (a) the

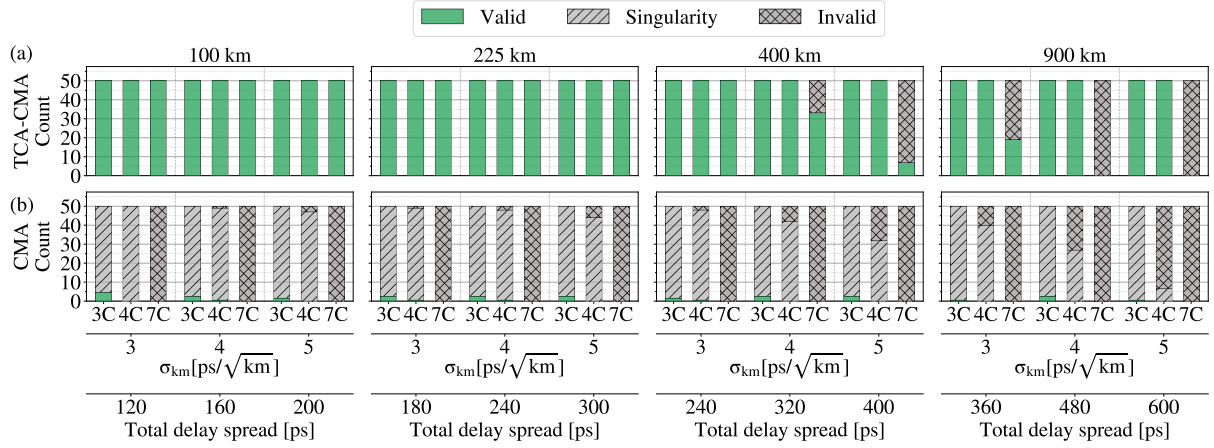


Fig. 2: Convergence outcomes: (a) TCA-CMA and (b) CMA evaluated on 3-, 4-, and 7-core RC-MCFs for various total delay spreads and distances. *Valid*: all $2N$ polarizations recovered without singularities; *singularity*: all $2N$ polarizations recovered but at least two are identical sequences; *invalid*: equalizer fails to recover $2N$ polarizations. 50 realizations per configuration.

TCA-CMA and (b) the conventional CMA ($\alpha = 0$) as pre-convergence equalizers. The conventional CMA configuration used identical equalizer parameters and pre-convergence iterations as the TCA-CMA. Convergence was verified through data-aided synchronization and the outcomes were classified as: *valid*, if all $2N$ polarizations converge to distinct sequences with correct time delays; *singularity*, if all $2N$ polarizations converge but at least two are identical sequences (or delayed replicas); and *invalid*, if fewer than $2N$ polarizations converge. The TCA-CMA (Fig. 2(a)) achieved singularity-free convergence for all investigated 3- and 4-core RC-MCF cases up to 900 km with $5 \text{ ps}/\sqrt{\text{km}}$ SMD, approximately 600 ps total delay spread (close to the 585 ps equalizer length). For 7-core RC-MCFs, convergence was ensured up to 225 km at $5 \text{ ps}/\sqrt{\text{km}}$ SMD (300 ps total delay spread); however, successful convergence was also observed at 400 km with $3 \text{ ps}/\sqrt{\text{km}}$ SMD (240 ps total delay spread), confirming that the total delay spread, not the SMD or distance, determines convergence. Notably, TCA-CMA never exhibited singularities; outcomes were exclusively valid or invalid, confirming the effectiveness of the tap-correlation penalty, as long as the equalizer converged. In contrast, the conventional CMA (Fig. 2(b)) failed in 98% of cases, predominantly converging to singularities for 3- and 4-core RC-MCFs and failing entirely for 7-core configurations. The results indicate that convergence becomes more challenging with higher core counts, even at total delay spreads well within the equalizer length. Enhanced initialization techniques or additional pre-convergence time may address this limitation.

We also evaluated SNR performance at 100 km and $5 \text{ ps}/\sqrt{\text{km}}$ SMD (Fig. 3). AWGN was added to simulate SNR_{ch} values of 10, 15, 20, and 25 dB, each with 50 realizations. The SNR was computed over the final 10^5 symbols of the MMA stage to exclude convergence phases of the MMA and

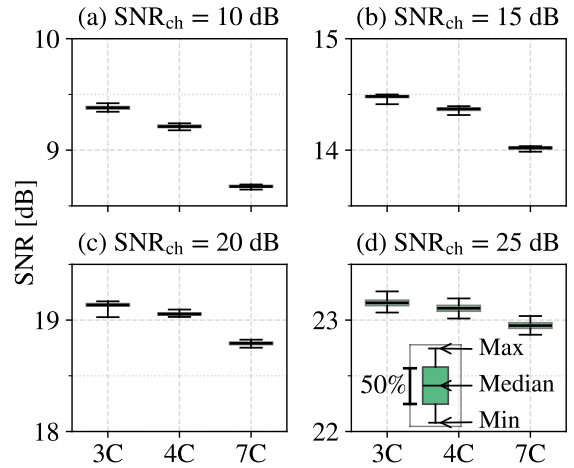


Fig. 3: Mean SNR (averaged over $2N$ polarizations) after MMA with TCA-CMA pre-convergence for 3-, 4-, and 7-core RC-MCFs under varying channel SNR (50 realizations each) at 100 km and $5 \text{ ps}/\sqrt{\text{km}}$ SMD (200 ps total delay spread).

CPE. For 3- and 4-core fibers, penalties remained below ≈ 1 dB up to 20 dB and below ≈ 2 dB at 25 dB. The 7-core configuration showed slightly higher penalties of up to ≈ 2 dB at 25 dB. As simulations without coupling and GD showed similar penalty increases with core count, we primarily attribute these variations to the non-data-aided MIMO equalizer's behavior at high core counts rather than to the impact of the coupling itself.

These results demonstrate that the TCA-CMA enables singularity-free MIMO equalization across various RC-MCF configurations, with the potential to reduce DSP complexity in SDM systems.

Conclusion

The proposed TCA-CMA achieves singularity-free MIMO equalization in RC-MCF systems by penalizing tap-weight correlations and is designed to reduce complexity compared to previous approaches. Simulations confirmed convergence up to 600 ps delay spread (3-, 4-cores) and 300 ps (7-cores). Results are promising for further studies, including a detailed parameter investigation, experimental validation, and real-time applications.

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